Multi-Agent Education System

Course Name: Agentic AI

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

Sr.no	Student Name	Enrolment Number
1	ABHIKET THAKUR	EN22CS301030
2	ADITYA PALIWAL	EN22CS301055
3	AARUSHI MITTAL	EN24CS3T10026
4	AKASH KUSHWAHA	EN22CS301083
5	AKSHAY DASHORE	EN22CS301096
6	AKSHITA RATHORE	EN22CS301100

Institution Name: Medi-caps University – Datagami Skill Based Course

Group: 04D1

Project No: AAI-04

Industry Mentor Name: Aashruti Shah

University Mentor Name: Prof. Nishant Shrivastava

Department of Computer Science & Engineering

Faculty of Engineering

MEDICAPS UNIVERSITY, INDORE- 453331

Jan - May 2026

1. Problem Statement & Objectives

1.1 Problem Statement

In today's digital age, students have access to a vast amount of information across various platforms. However, this abundance of information often leads to challenges such as information overload, lack of structure, and difficulty in identifying reliable and relevant content for effective learning. Students are required to spend significant time searching, filtering, and organizing study material from multiple sources, which reduces learning efficiency. Existing educational tools primarily provide raw information or static content and do not offer intelligent assistance in organizing knowledge into structured and easy-to-understand formats. Additionally, most systems lack collaborative intelligence and fail to simulate the way experts approach learning—by researching, organizing, and presenting information systematically.

There is a need for an intelligent system that can automate the process of research and content generation, providing well-structured study material tailored to a given topic. Such a system should minimize manual effort, improve learning efficiency, and deliver high-quality educational content in a simplified format. To address this problem, the Multi-Agent Education System leverages a multi-agent architecture using CrewAI and integrates advanced language models via Groq to automatically generate structured study guides. The system aims to enhance self-learning by combining research and content creation capabilities into a unified, intelligent workflow.

1.2 Project Objectives

The primary objective of the Multi-Agent Education System is to develop an intelligent and automated platform that generates structured and high-quality educational content for any given topic. The system aims to simplify the learning process by reducing the effort required to search, organize, and understand information.

The specific objectives of the project are as follows:

- To design a system that can automatically generate comprehensive study material for a user-defined topic without manual research.
- To utilize a collaborative agent-based approach using CrewAI, where different agents perform specialized tasks such as research and content writing.
- To ensure that the generated content is well-organized, readable, and suitable for effective learning, including key concepts, explanations, and examples.
- To leverage high-performance language models through Groq to generate accurate and meaningful educational content.
- To create a simple and interactive interface using Streamlit that allows users to easily input topics and access results.
- To enable saving of generated study guides in a structured format for future reference and reuse.

- To design the system in a modular way so that additional agents or features can be added in the future without major changes to the existing system.
- To Show how multi-agent systems and AI technologies can be effectively applied to improve learning and knowledge accessibility.

1.3 Scope of the Project

The Multi-Agent Education System is designed to provide an intelligent solution for generating structured educational content based on user-defined topics. The system focuses on simplifying the learning process by automating research and content organization using a multi-agent approach.

The scope of the project includes developing a web-based application using Streamlit that allows users to input a topic and receive a well-structured study guide. The system utilizes a multi-agent architecture implemented through CrewAI, where different agents are responsible for specific tasks such as gathering information and generating readable content. The application integrates advanced language models via Groq to ensure fast and effective content generation.The system supports the generation of study material that includes explanations, key concepts, and examples, and stores the output in markdown format for future reference. It is designed with a modular structure, enabling easy scalability and the addition of new features or agents.

However, the current scope is limited to offline AI-generated content and does not include features such as real-time data retrieval, user authentication, personalized learning, or advanced assessment tools like quizzes and progress tracking. The system also operates with a simplified storage mechanism without the use of databases.The project can be further extended by incorporating features such as personalized recommendations, integration with learning platforms, multi-language support, and additional agents for quizzes, flashcards, or interactive learning.

2. Proposed Solution

The proposed solution is to develop a Multi-Agent Education System that automates the generation of structured study material using a collaborative AI-based approach. The system utilizes multiple agents coordinated through CrewAI, where each agent performs a specific task such as research and content creation. A user-friendly interface built with Streamlit allows users to input topics, while advanced language models integrated via Groq ensure fast and high-quality content generation.

2.1 Key features

The Multi-Agent Education System includes several important features that enhance its functionality and usability:

- **Automated Study Material Generation:** The system automatically generates structured and comprehensive study guides for any user-defined topic, reducing the need for manual research.
- **Multi-Agent Architecture:** Implements a collaborative approach using CrewAI, where different agents perform specialized tasks such as research and content writing.

- **Structured and Readable Output:** The generated content is well-organized into sections, including explanations, key concepts, and examples, making it easy to understand and use for learning.
- **Fast AI-Powered Processing:** Utilizes advanced language models through Groq to ensure quick and efficient content generation.
- **User-Friendly Interface:** Provides a simple and interactive interface built using Streamlit, allowing users to easily input topics and view results.
- **Modular and Scalable Design:** The system is designed with modular components, enabling easy addition of new agents or features without affecting existing functionality.
- **File-Based Output Storage:** Generated study guides are saved in markdown format, allowing users to access, store, and reuse the content easily.
- **Efficient Data Flow and Processing:** Ensures smooth flow of data between components, from user input to final output, using a well-defined process pipeline.

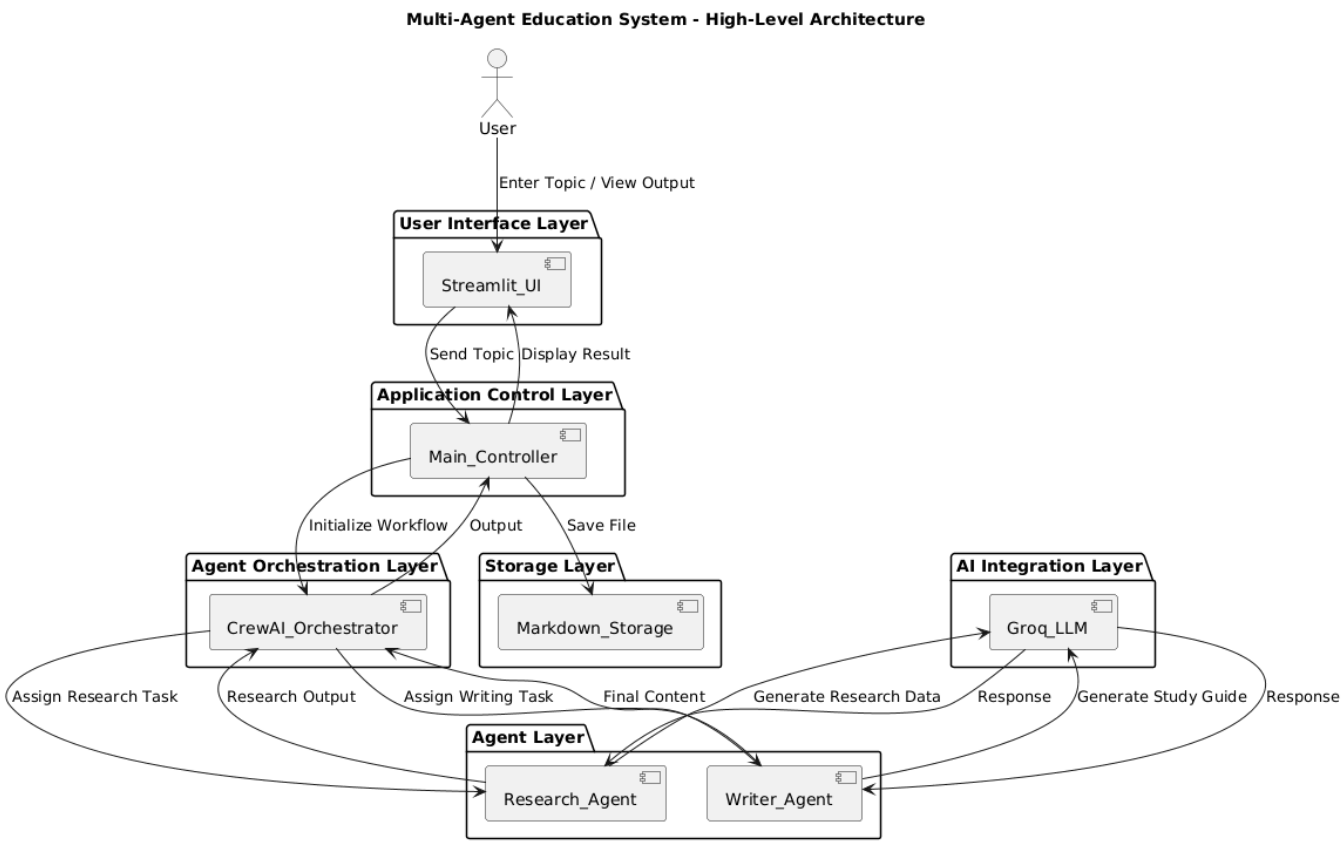
2.2 Overall Architecture / Workflow

The Overall Architecture of the Multi-Agent Education System defines how different components interact to generate structured study material. The system follows a sequential workflow where user input is processed through a series of modules, including the user interface, control logic, and a multi-agent framework implemented using CrewAI. Each agent performs a specific task, such as research and content generation, and the final output is produced efficiently using AI models integrated via Groq. This workflow ensures smooth data flow, modularity, and effective content generation.

High-Level Architecture

The system consists of the following major components:

1. **User Interface (Frontend) –** Developed using Streamlit, it accepts the topic input from the user and displays the generated study material in a structured format.
2. **Application Control Layer (Backend Controller) –** Handles user requests, validates input, manages the overall workflow, and coordinates communication between different components.
3. **Agent Orchestration Layer –** Implemented using CrewAI, it manages multiple agents and ensures sequential execution of tasks.
4. **Research Agent –** Responsible for gathering relevant information, identifying key concepts, and preparing structured research data for the given topic.
5. **Writer Agent –** Converts the research data into a well-organized and easy-to-understand study guide with proper structure and explanations.
6. **Language Model Integration (LLM) –** Powered by Groq, it enables intelligent content generation and ensures fast and accurate responses.
7. **Storage Module –** Saves the generated study guides in markdown format within the local file system for future access and reuse.



System Workflow

Step 1: User Input

- The user enters a topic through the web interface developed using Streamlit.
- The input is submitted for processing.

Step 2: Input Processing

- The Application Controller receives the user input.
- The input is validated and preprocessed to ensure correctness.
- The processed input is forwarded to the orchestration layer.

Step 3: Task Initialization

- The Agent Orchestration Layer implemented using CrewAI initializes the workflow.
- The task is assigned to the Research Agent.

Step 4: Research Phase

- The Research Agent interacts with the language model integrated via Groq.
- Relevant information and key concepts related to the topic are gathered.
- The research output is structured and sent back to the orchestrator.

Step 5: Content Generation

- The orchestrator assigns the task to the Writer Agent.
- The Writer Agent uses the research data to generate structured educational content.
- The content includes explanations, key points, and examples.

Step 6: Output Processing

- The generated content is returned to the Application Controller.
- The content is formatted into a readable structure (markdown format).

Step 7: Storage and Display

- The final study material is stored in the local storage system.
- The output is displayed to the user through the interface.

Information Movement Across Layers:

User Layer → Presentation Layer

- User submits the topic or prompt
- Input is captured through the Streamlit interface

Presentation Layer → Application Layer

- Input is validated and preprocessed
- Cleaned input is forwarded to the controller

Application Layer → Orchestration Layer

- Task is initialized and managed
- Workflow is assigned to agents

Orchestration Layer → Agent Layer

- Research Agent gathers information
- Writer Agent generates structured content

Agent Layer → AI Integration Layer

- Agents interact with the language model via Groq
- AI generates responses based on given tasks

Agent Layer → Application Layer

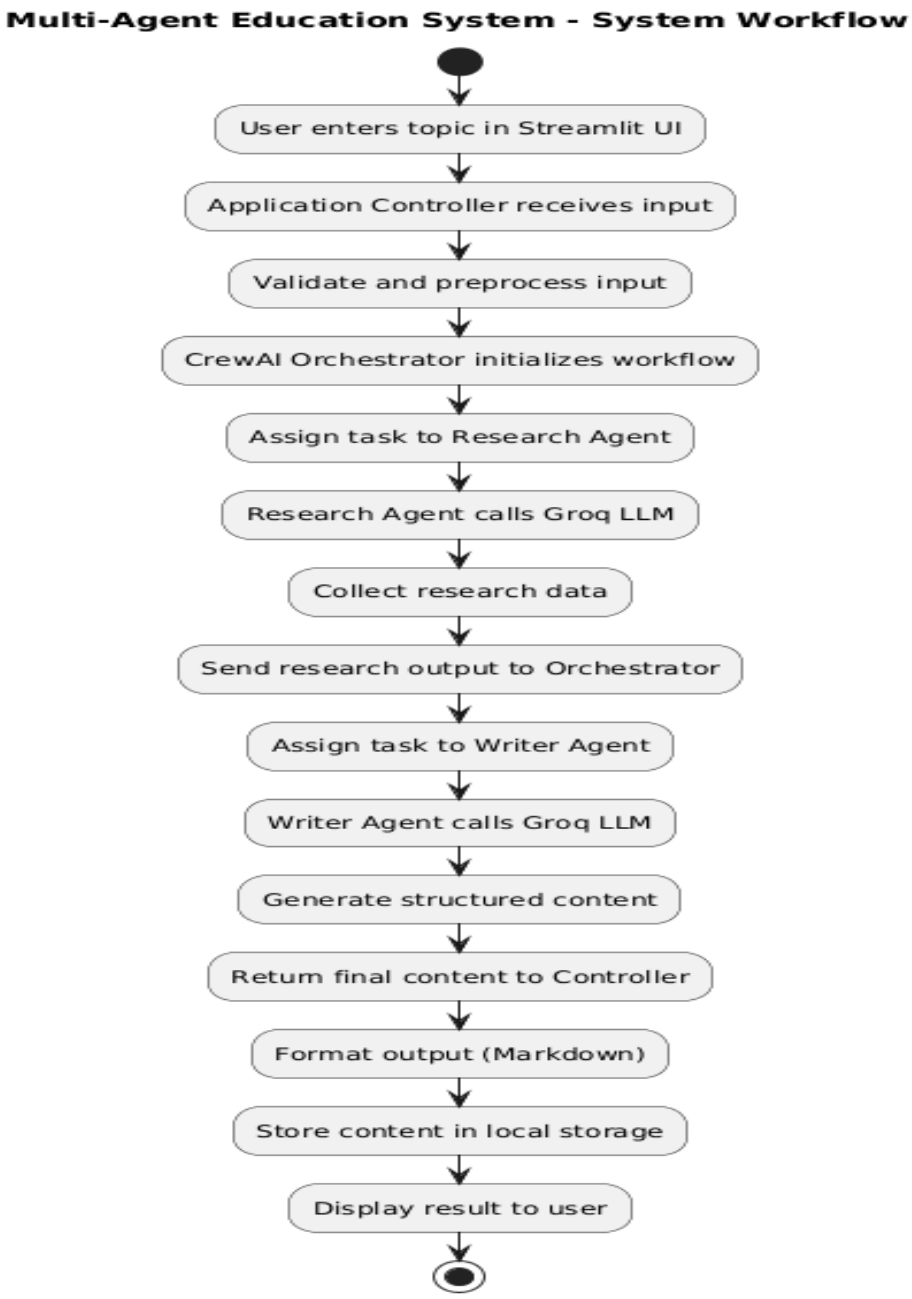
- Generated content is returned to the controller
- Output is structured and formatted

Application Layer → Presentation Layer

- Final response is sent to the frontend
- Content is displayed to the user

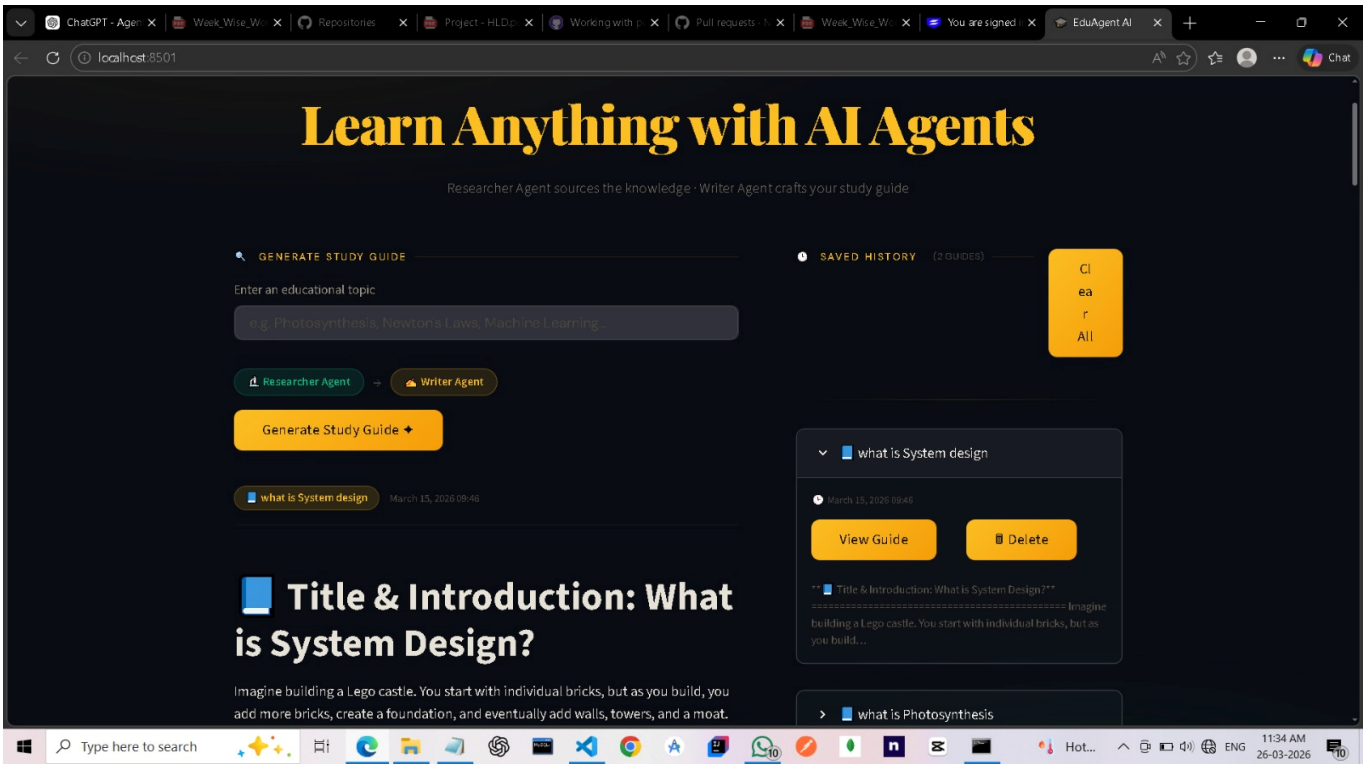
Application Layer → Storage Layer

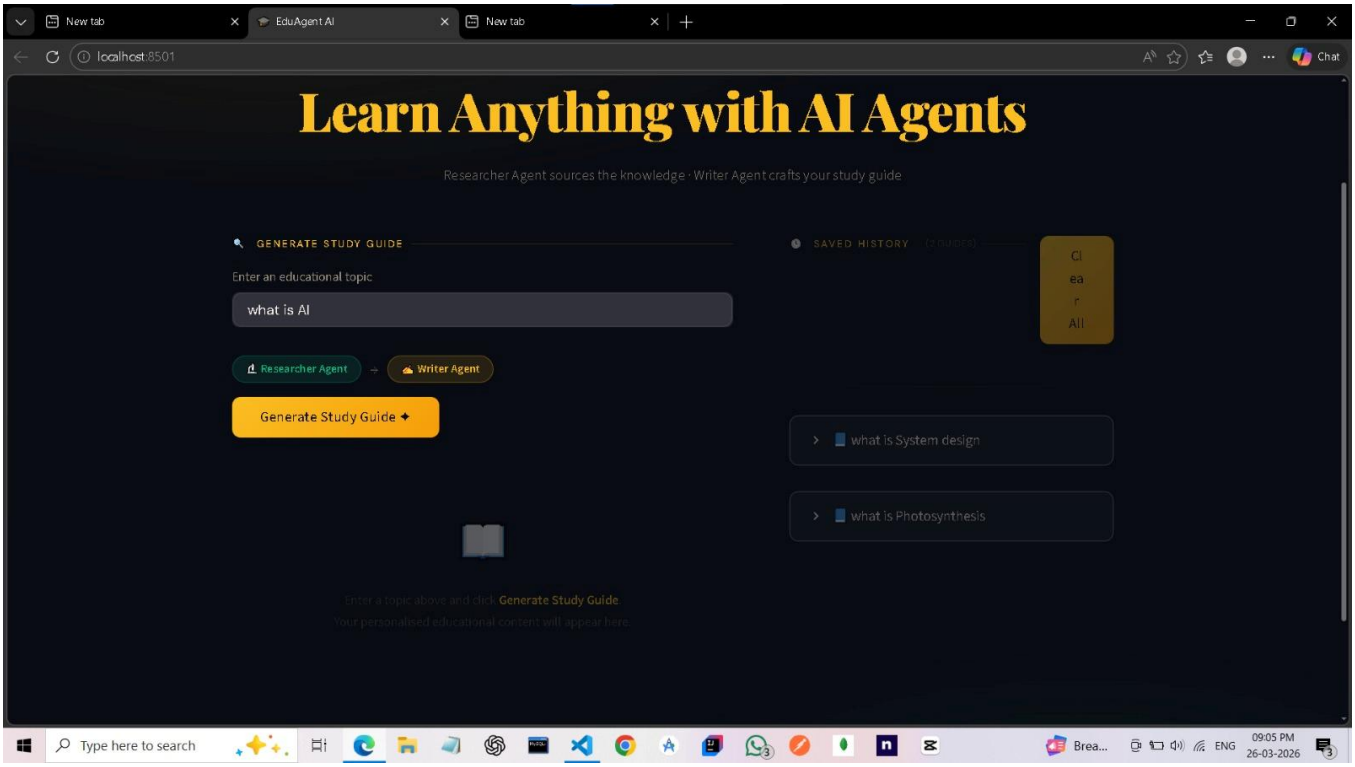
- Generated study material is saved in markdown format

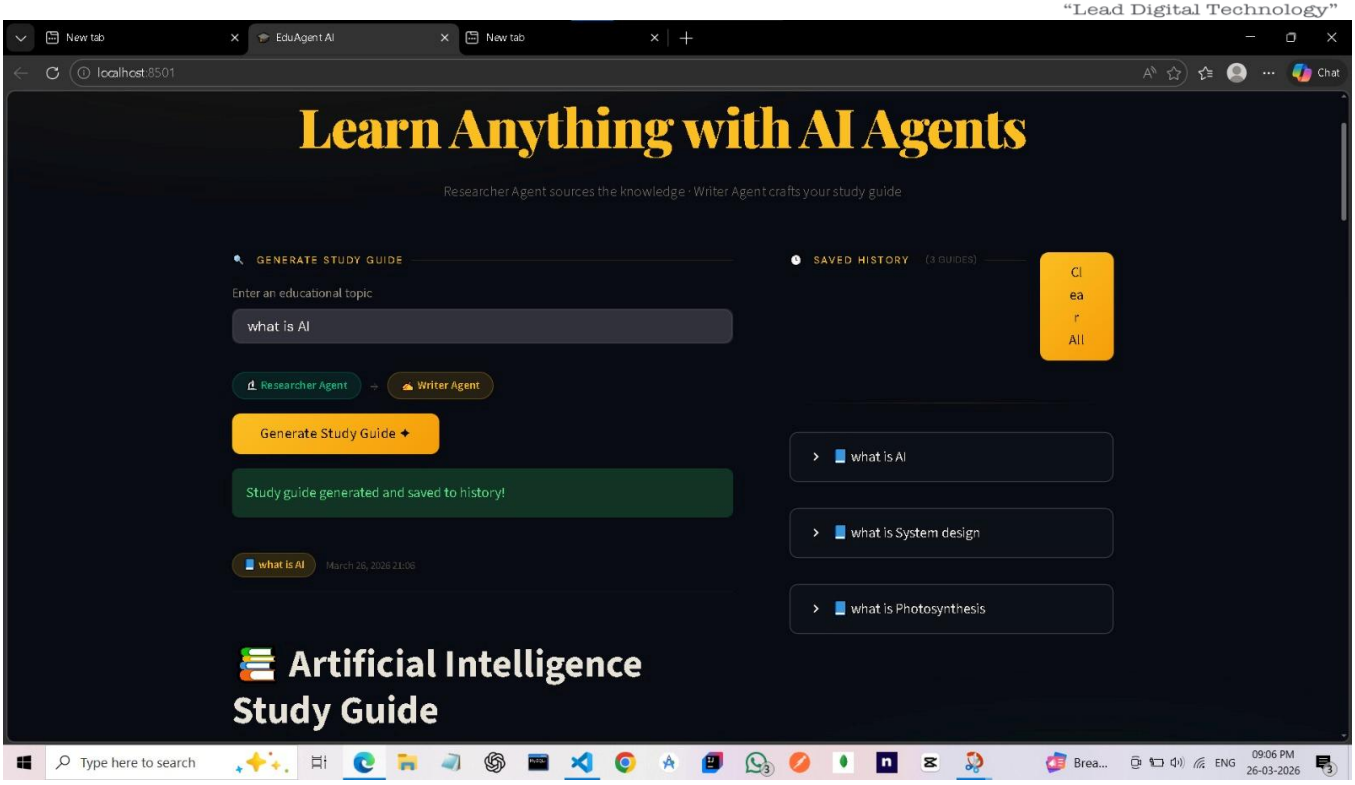


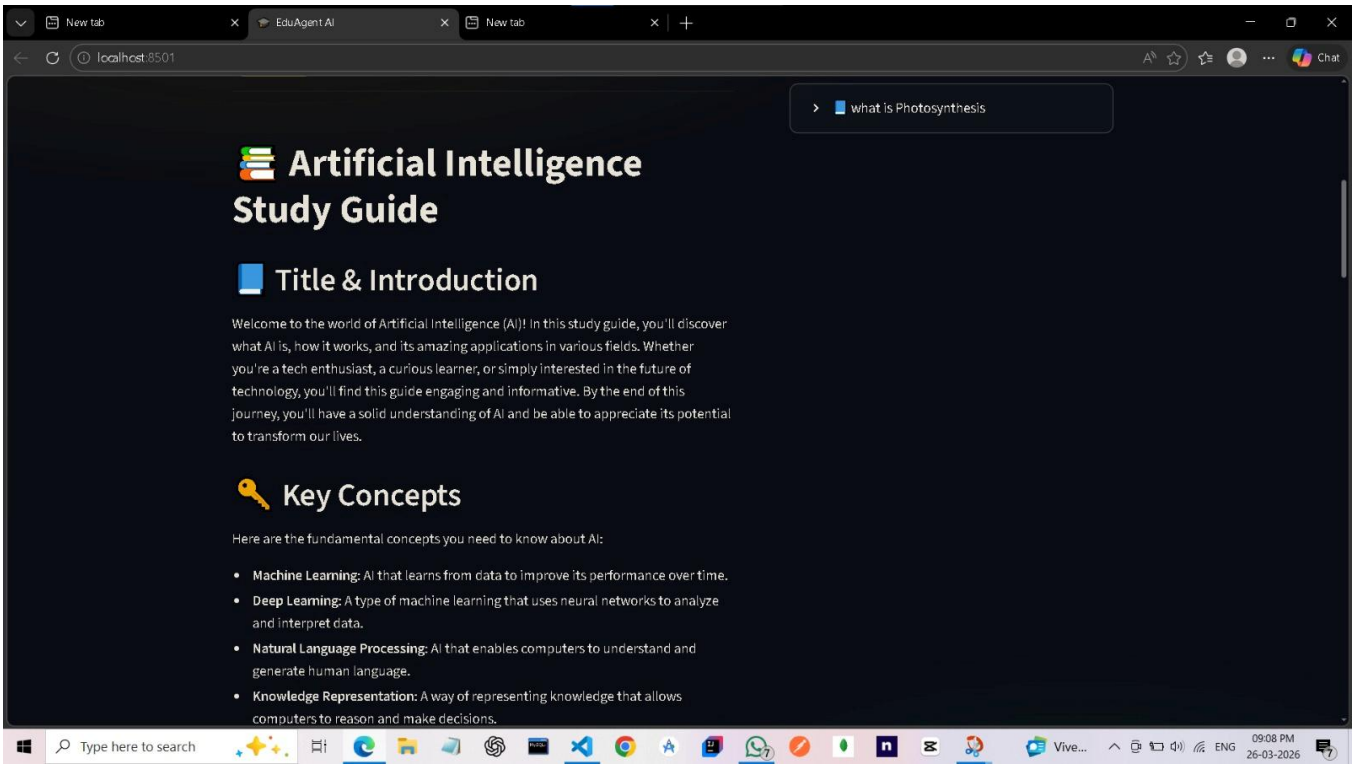
3. Results & Output

3.1 Screenshots / outputs









3.2 Reports / Dashboards / Models

This section presents the outputs and key components of the Multi-Agent Education System, including the generated study materials, system interface, and underlying AI models. The application provides structured reports in the form of markdown-based study guides, while the user interface developed using Streamlit acts as a simple dashboard for interaction. Additionally, the system leverages advanced language models integrated via Groq and coordinated through CrewAI to ensure efficient and high-quality content generation.

AI Models Used

- **Language Model (LLM)** – The core model integrated via Groq is responsible for understanding user input and generating structured educational content. It performs tasks such as concept explanation, content generation, and contextual reasoning.
- **Research Processing (LLM Usage)** – The Research Agent uses the language model to gather relevant information, extract key concepts, and organize topic-related knowledge.
- **Content Generation Model (LLM Usage)** – The Writer Agent uses the same language model to transform research data into a well-structured study guide with clear explanations and examples.
- **Multi-Agent Coordination Model** – Implemented using CrewAI, this is not a learning model but a framework that manages task distribution, sequencing, and collaboration between agents.
- **Natural Language Understanding (NLU) Capability** – Enables the system to understand user input, context, and intent to generate meaningful and accurate responses.
- **Contextual Reasoning Model** – Helps maintain logical flow and coherence in the generated content across different sections.

System Reports

The system can generate or maintain basic reports such as:

- **User Query Logs** – Records the topics entered by users, helping to identify frequently searched subjects and usage patterns.
- **Content Generation Logs** – Maintains logs of generated study materials, including timestamps and processed topics.
- **Agent Execution Logs** – Tracks the execution of different agents (Research Agent and Writer Agent) managed by CrewAI, including task completion status.
- **Response Time Monitoring** – Measures the time taken to generate study material, helping evaluate system performance.
- **Error and Exception Logs** – Captures system errors or failures during processing to assist in debugging and system improvement.
- **Output Storage Records** – Keeps track of saved markdown files, including file names and storage details.
- **API/Model Usage Logs** – Monitors interactions with the language model integrated via Groq, including request frequency and usage patterns.

Dashboard (Optional Enhancement)
A monitoring dashboard can be implemented to display:

- **Total Number of Queries Processed** – Shows how many topics have been submitted and processed by the system.

- **Most Searched Topics** – Highlights frequently requested topics to understand user interests and trends.
- **Average Response Time** – Displays the average time taken to generate study material, helping evaluate system performance.
- **Agent Execution Status** – Provides real-time status of agents (Research and Writer) managed by CrewAI.
- **System Health Status** – Indicates whether the system is functioning properly, including API availability and processing status.
- **Content Generation Count** – Tracks the number of study guides successfully generated and stored.
- **Model/API Usage Overview** – Displays usage statistics of the language model integrated via Groq, such as request count and activity levels.
- **Storage Overview** – Shows details of saved markdown files, including count and recent outputs.

The dashboard provides real-time insights into system performance and usage trends.

3.3 Key outcomes

The successful completion of this project resulted in the following key outcomes:

- Successfully developed a Multi-Agent Education System capable of generating structured and meaningful study material based on user-defined topics.
- Implemented a multi-agent architecture using CrewAI, enabling task specialization through Research and Writer agents.
- Achieved efficient and high-quality content generation by integrating advanced language models via Groq.
- Designed a simple and interactive user interface using Streamlit, allowing users to easily input topics and view results.
- Enabled automatic generation of well-structured educational content, including explanations, key concepts, and examples.
- Implemented a modular and scalable system design, allowing future enhancements such as adding new agents or features.
  Ensured fast response time and smooth workflow execution through efficient coordination between system components.
- Provided a lightweight storage solution by saving generated content in markdown format for future reference.
- Demonstrated the practical application of AI and multi-agent systems in education, showcasing how automation can simplify learning processes.

4. Conclusion

The Multi-Agent Education System successfully demonstrates how artificial intelligence can be leveraged to automate the generation of structured educational content. By utilizing a multi-agent architecture implemented through CrewAI, the system efficiently divides tasks between specialized agents, ensuring better organization and quality of output.

The integration of advanced language models via Groq enables fast and accurate content generation, while the user-friendly interface developed using Streamlit ensures ease of use. The system effectively transforms a simple user input into a comprehensive study guide, including key concepts and explanations.

Overall, the project highlights the potential of multi-agent systems in the field of education by simplifying content creation and improving accessibility to learning resources. Although the current implementation is limited in terms of personalization and advanced features, it provides a strong foundation for future enhancements such as adaptive learning, real-time data integration, and interactive learning tools.

5. Future Scope & Enhancements

The current Multi-Agent Education System provides a strong foundation for automated content generation; however, it can be further enhanced with several advanced features to improve functionality, scalability, and user experience.

- **Personalized Learning:** The system can be extended to include user profiles and learning preferences, enabling customized content generation based on individual learning styles and progress.
- **User Authentication & Dashboard:** Adding authentication features would allow users to save, manage, and revisit previously generated study materials through a dedicated dashboard.
- **Integration with External Knowledge Sources:** Incorporating real-time data retrieval from external APIs, research databases, or the web can improve the accuracy and relevance of generated content.
- **Addition of More Specialized Agents:** The system can be enhanced by introducing new agents (via CrewAI) such as:
    - Quiz Generator Agent
    - Flashcard Creator Agent
    - Summarization Agent
    - Doubt-Solving Agent
- **Multi-Language Support:** Enabling support for multiple languages would make the system accessible to a broader audience and improve inclusivity.

- **Advanced AI Capabilities:** Leveraging more advanced models via Groq can further improve content quality, reasoning, and contextual understanding.
- **Database Integration:** Replacing the current file-based storage with a database system would improve scalability, data management, and retrieval efficiency.
- **Interactive Learning Features:** Features such as quizzes, progress tracking, and performance analytics can be added to make the learning experience more engaging and effective.
- **Mobile and Cross-Platform Support:** Developing mobile or cross-platform applications can enhance accessibility and usability for users on different devices.
- **Collaboration and Sharing Features**: Users could be allowed to share study materials or collaborate with others, making the platform more interactive and community-driven.